# Exam 3 Review

Intro to Audio Tech 1
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Monday, November 24

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## Lesson Outline

- Exam Review
- Final Project Discussion
- Up Next…

# Exam Format

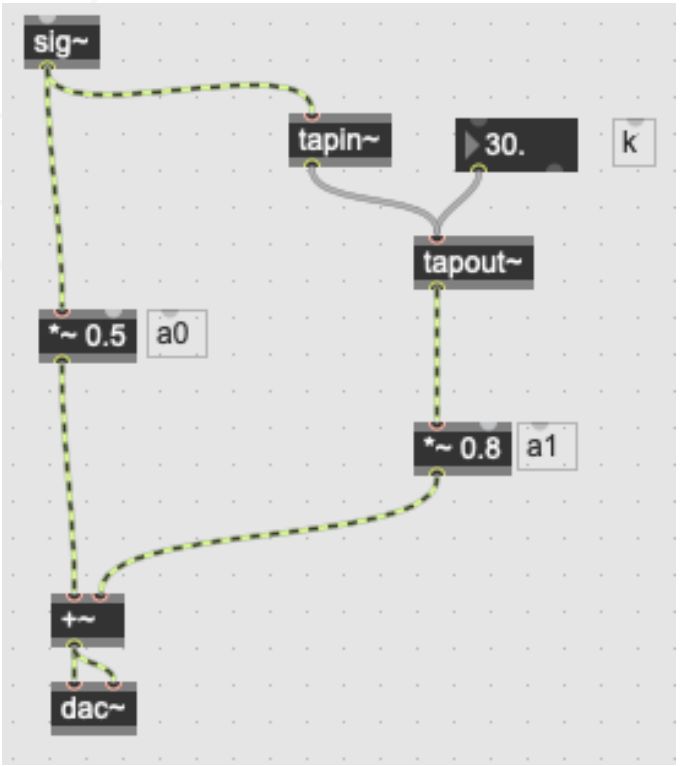
- PDF download
- Handwrite answers, then upload
- Available this evening through next Monday at 11:59p
- Question types:
  - True/False
  - Multiple Choice
  - Short answer – math and explanation
  - Build a Max patch

# Exam Topics

- Phasors and Delays
  - Euler's identity
  - Sinusoids, delays, notation (DSP First Chapter)
  - Graphic notation
- Filters
  - Filter representations
    - Block diagram, difference equation, phasor
  - Frequency response
    - Magnitude and phase response
  - Comb and Allpass filters
- Delay Based effects
  - Flanger, Chorus, Delay, Phaser
  - Delay lines in Max
  - Understand filtering/phase shift effects
- Fourier Transform
  - Process of Fourier Series, FT, DFT, STFT
  - Moving between time and frequency domain
    - Windowing, periodization
  - Relationship to SR
  - Tradeoff time vs frequency resolution

# Quiz 3

# Rewriting this “Filter”




Difference equation:

$$y_t = a0 * x_t + a1 * x_{t-k}$$

Phasor notation:

$$y_t = a0 * e^{j\omega t} + a1 * e^{j\omega(t-k)}$$

Frequency response:

$$y_t = a0 * e^{j\omega t} + a1 * e^{j\omega(t-k)}$$

$$e^{j\omega t}(a0 + a1 * e^{-j\omega k})$$

$$H(\omega) = a0 + a1 * e^{-j\omega k}$$

# IR and Filter Input

Difference equation:

$$y[n] = x[n] + 2x[n-1] + 0.5x[n-2]$$

Impulse response:

$$\delta[n] = [1, 2, 0.5]$$

Calculate first 5 samples of x[n] = [0, 2, 4, 3, 1]:

$$y[0] = x[0] + 2x[-1] + 0.5x[-2] = 0$$
$$y[1] = x[1] + 2x[0] + 0.5x[-1] = 2 + 0 = 2$$
$$y[2] = x[2] + 2x[1] + 0.5x[0] = 4 + 2(2) + 0 = 8$$
$$y[3] = x[3] + 2x[2] + 0.5x[1] = 3 + 2(4) + 0.5(2) = 12$$
$$y[4] = x[4] + 2x[3] + 0.5x[2] = 1 + 2(3) + 0.5(4) = 9$$
$$y[n] = [0, 2, 8, 12, 9]$$

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# Same question using convolution

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

*

| | | |
|---|---|---|
| 1 | 2 | 0.5 |

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

| | | |
|---|---|---|
| 0,5 | 2 | 1 |

0(1) = 0

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | | | | | | |

# Same question using convolution

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

*

| | | |
|---|---|---|
| 1 | 2 | 0.5 |

| | | | | | |
|---|---|---|---|---|---|
| | 0 | 2 | 4 | 3 | 1 |
| 0.5 | 2 | 1 | | | |

0(1) = 0

0(2) + 2(1) = 2

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 2 | | | | | |

# Same question using convolution

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

*

| | | |
|---|---|---|
| 1 | 2 | 0.5 |

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

| | | |
|---|---|---|
| 0.5 | 2 | 1 |

0(1) = 0

0(2) + 2(1) = 2

0(0.5) + 2(2) + 4(1) = 8

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 2 | 8 | | | | |

# Same question using convolution

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |

*

| | | |
|---|---|---|
| 1 | 2 | 0.5 |

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 4 | 3 | 1 |
| | | 0.5 | 2 | 1 |

0(1) = 0

0(2) + 2(1) = 2

0(0.5) + 2(2) + 4(1) = 8

2(0.5) + 4(2) + 3(1) = 12

4(0.5) + 3(2) + 1(1) = 9

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 2 | 8 | 12 | 9 | | |

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**A complex sinusoid x[n] has frequency 11025 Hz, amplitude 50, and initial phase of 135 degrees. Another complex sinusoid, y[n], has the same frequency but an amplitude 20 and initial phase 45 degrees. What is resultant phasor of the sum of x[n] and y[n]? (2 points)**

- Phasor x[n]
  - $50\angle 135$
  - $50\cos(135) + 50jsin(135) = 50\left(-\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right)$
- Phasor y[n]
  - $20\angle 45$
  - $20\cos(45) + 20jsin(45) = 20\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right)$
- Sum (cartesian)
  - Real: $50\left(-\frac{\sqrt{2}}{2}\right) + 20\left(\frac{\sqrt{2}}{2}\right) = -21.21$
  - Imag: $50\left(j\frac{\sqrt{2}}{2}\right) + 20\left(j\frac{\sqrt{2}}{2}\right) = 49.5j$
- Polar
  - Magnitude -> $\sqrt{-21.21^2 + 49.5^2} = 53.85$
  - Phase -> arctan(49.5/21.21) = 66.8 but its in Quad 2 so 180-66.8 = 113.2
  - $53.85\angle 113$

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# Exam 3 Review

# Questions to consider

- What are the similarities and differences between a comb filter and a flanger?
- How do you build a chorus effect?
- How does a large vs small feedback gain effect a delay-line system?
- Why use allpass and comb filters to create a reverberator?

## Match the Frequency Response to the Difference Equation

- A. $H(\omega) = e^{-j\omega} + e^{-j3\omega}$
- B. $H(\omega) = 1 - e^{-j2\omega}$
- C. $H(\omega) = e^{-j2\omega}(2jsin(\omega))$

- 1. $y_n = x_{n-1} - x_{n-3}$
- 2. $y_n = x_{n-1} + x_{n-3}$
- 3. $y_n = x_n - x_{n-2}$

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